

PCA REGIME SIGNAL

MNQ Futures Trading Strategy

Based on Infantino & Itzhaki — MIT Sloan Management School, 2010

"Developing High-Frequency Equities Trading Models"

1. Strategy Overview

This indicator applies Principal Component Analysis (PCA) — a technique from quantitative finance — to a basket of five futures contracts that proxy the major risk factors driving Nasdaq futures (MNQ). The strategy detects whether the market is in a **momentum regime** or a **mean-reversion regime**, then trades accordingly when MNQ drifts far enough from the predicted level implied by those factors.

All five factor proxies trade the 23-hour CME/CBOT/COMEX session, so signals are available overnight alongside MNQ — unlike ETF-based approaches that go stale after the 4 pm US equity close.

Factor Proxies

PC	Symbol	Exchange	What It Captures
PC1	ES1!	CME Mini	Broad market beta / S&P 500 direction
PC2	YM1!	CBOT Mini	Value / cyclical tilt (Dow Jones)
PC3	RTY1!	CME Mini	Risk appetite / small-cap premium
PC4	ZN1!	CBOT	Rate / duration sensitivity (10-yr T-Note)
PC5	GC1!	COMEX	Risk-off / USD strength (Gold)

2. Core Mathematics

Step 1 — Log Returns

At every bar, compute the log return for MNQ and each of the five factor proxies:

$$r_j(t) = \ln(P_j(t) / P_j(t-1))$$

Log returns are used instead of raw price changes because they are additive over time and symmetrical around zero — properties that make the regression and volatility math well-behaved.

Step 2 — Rolling OLS Betas

For each factor j , estimate how sensitive MNQ's return is to that factor over a rolling window of **reg_len** bars (default 100):

$$\text{beta}_j = \text{corr}(r_{\text{MNQ}}, r_j) \times (\text{stdev}(r_{\text{MNQ}}) / \text{stdev}(r_j))$$

This is the standard OLS slope formula expressed through correlation. A beta near +1 for ES1! means MNQ has moved almost in lockstep with the S&P. A negative beta for ZN1! (bonds) means MNQ tends to fall when rates rally — the classic equity/bond flight-to-safety dynamic.

Step 3 — Factor Contributions

Multiply each factor's return by its beta to get its contribution to MNQ's predicted move at bar t :

$$d_j(t) = \text{beta}_j(t) \times r_j(t)$$

Think of d_j as: 'how many points of MNQ return does factor j explain right now?'

Step 4 — Cross-Sectional Volatility (sigma_D)

Compute the standard deviation of the five contributions {d_1 ... d_5} at each bar:

$$\text{sigma_D}(t) = \text{stdev}(d_1, d_2, d_3, d_4, d_5)$$

When sigma_D is **high**, the five factors are pulling in very different directions — the market is fragmented and noisy. When sigma_D is **low**, all factors move together — the market is in a coherent, trending state. This single number is the heartbeat of the regime detector.

Step 5 — Psi: Rate of Change of sigma_D

Take the first difference of sigma_D:

$$\text{psi}(t) = \text{sigma_D}(t) - \text{sigma_D}(t-1)$$

Psi measures how quickly factor dispersion is changing. A rising psi means factors are increasingly diverging; a falling psi means they are converging.

Step 6 — E_H: The Regime Detector

Compute the L2 norm (Euclidean distance) of psi over the last H bars (default 20):

$$E_H(t) = \sqrt{\sum_{i=0}^{H-1} \text{psi}(t-i)^2}$$

E_H answers: 'how much has factor dispersion been churning over the last H bars?' This is the key insight from Infantino & Itzhaki:

E_H State	Regime	Interpretation	Trade Bias
E_H rising (E_H > E_H[1])	MOMENTUM	Dispersion is accelerating. Factors are increasingly out of sync.	Trade WITH the drift (follow the dislocation)
E_H falling (E_H < E_H[1])	MEAN REVERSION	Dispersion is stabilising. Factors are converging back together.	Trade AGAINST the drift (fade the dislocation)

3. Dislocation Signal

Even knowing the regime, we need to know *when* to trade. The dislocation answers that question by measuring how far MNQ has wandered from where the five factors say it should be.

Predicted Cumulative Return

Sum each factor contribution over the last H bars:

$$\text{predicted_cum_ret} = \sum_{i=0}^{H-1} (\text{beta}_j \times d_j(t-i)) \text{ for } j=1..5$$

Dislocation

Subtract the predicted cumulative return from MNQ's actual cumulative return over the same H bars:

$$\text{dislocation}(t) = \text{actual_cum_ret}(t) - \text{predicted_cum_ret}(t)$$

A **positive dislocation** means MNQ has outrun what the factors predict — it looks expensive relative to its drivers. A **negative dislocation** means MNQ has lagged — it looks cheap.

Signal Filter

To avoid trading every small wiggle, a signal is only considered strong enough when the dislocation exceeds `sig_mult` standard deviations of its own recent history (default: 1.5×):

```
signal_strong = |dislocation(t)| > sig_mult x stdev(dislocation,  
reg_len)
```

4. Signal Generation Logic

The raw signal and regime are combined into a single trade direction. The regime flips whether we fade or follow the dislocation:

Dislocation	Raw Signal	Regime	Final Trade Signal
Positive (MNQ expensive)	SHORT	MEAN REVERSION	SHORT — fade the gap
Positive (MNQ expensive)	SHORT	MOMENTUM	LONG — follow the gap
Negative (MNQ cheap)	LONG	MEAN REVERSION	LONG — fade the gap
Negative (MNQ cheap)	LONG	MOMENTUM	SHORT — follow the gap

Anti-Whipsaw Confirmation

Because dislocation can oscillate around the threshold, the indicator requires **confirm_bars** consecutive bars of the same signal direction before entering (default: 2 bars). A streak counter tracks this:

```
sig_streak +=1 if signal unchanged, else reset to 1 confirmed_signal =  
signal if sig_streak >= confirm_bars, else FLAT
```

5. Trade Entry, TP and Stop Loss

Entry

A new trade is entered on the bar that the confirmed signal appears — at the closing price of that bar. Only one position is held at a time.

Stop Loss

The stop loss distance is ATR-based (default) or a fixed number of points:

```
SL_distance = ATR(14) x sl_atr_mult (default: 1.5x) SL_price = entry -  
SL_distance (long) SL_price = entry + SL_distance (short)
```

Using ATR makes the stop adaptive — wider in volatile sessions, tighter in calm ones. The current ATR(14) is shown in the stats table on the chart.

Take Profit

The take profit is a fixed multiple of the stop distance (default 2:1 R:R):

```
TP_distance = SL_distance x rr_ratio (default: 2.0) TP_price = entry +  
TP_distance (long) TP_price = entry - TP_distance (short)
```

Exit Rules

Exit Trigger	Condition	Label
Take Profit hit	High >= TP (long) or Low <= TP (short)	[v TP HIT]
Stop Loss hit	Low <= SL (long) or High >= SL (short)	[x SL STOP]
Signal reversal	confirmed_signal flips to opposite direction (after min_hold bars)	[EXIT]
Signal goes flat	confirmed_signal == 0 (after min_hold bars)	[EXIT]

min_hold prevents the exit logic from firing for the first N bars after entry (default 3). This avoids getting shaken out on the very next bar if the signal briefly dips below threshold.

6. Chart Visual Outputs

Visual Element	Color	Meaning
Chart background — teal	Teal	Mean-reversion regime (E_H falling)
Chart background — red	Red	Momentum regime (E_H rising)
Chart background — green	Green	Active long trade in progress
Chart background — red	Red	Active short trade in progress
Triangle UP (L)	Lime	Long entry — plotted below the candle
Triangle DOWN (S)	Red	Short entry — plotted above the candle
Yellow X	Yellow	Trade exit (any reason)
White dotted line	White	Entry price level — extends right
Red dashed line	Red	Stop loss level — extends right, deletes on exit
Green dashed line	Lime	Take profit level — extends right, deletes on exit
Label at entry arrow	Green / Red / Gray	Shows TP / Entry / SL prices; header updates to [v TP HIT], [x SL STOP], or

7. Parameter Reference

Parameter	Default	What It Controls
H	20	Lookback window for E_H and cumulative return calculation (bars)
reg_len	100	Rolling window for beta regression and dislocation stdev
sig_mult	1.5	Minimum dislocation size to generate a signal (in stdev units); 0 = trade every bar
confirm_bars	2	Consecutive bars of same signal required before entry
min_hold	3	Minimum bars to hold before an exit signal is honoured
sl_mode	ATR	Stop loss method: ATR-adaptive or Fixed Points
sl_atr_mult	1.5	Multiplier applied to ATR(14) to set the stop distance
sl_fixed_pts	20	Fixed stop distance in MNQ points (used when sl_mode = Fixed)
rr_ratio	2.0	Take-profit = stop distance x this ratio (2.0 = 2:1 R:R)
mnq_tick	0.25	MNQ tick size in points (for dollar P&L calculation)
mnq_dpt	0.50	Dollar value per tick (\$0.50 for MNQ)

8. Signal Flow Summary

Step	Calculation	Output
1	Log returns for MNQ and 5 proxies	r_0 ... r_5
2	Rolling OLS beta over reg_len bars	beta_1 ... beta_5
3	Factor contributions: $d_j = \text{beta}_j \times r_j$	d_1 ... d_5
4	Cross-sectional stdev of {d_1...d_5}	sigma_D
5	First difference: $\text{psi} = \text{sigma}_D - \text{sigma}_D[1]$	psi
6	L2 norm of psi over H bars	E_H
7	$E_H > E_H[1]$?	Regime: MOMENTUM or MEAN REV
8	Actual cum_ret minus predicted cum_ret over H bars	dislocation
9	$ \text{dislocation} > \text{sig_mult} \times \text{stdev}(\text{dislocation})$?	signal_strong = True/False
10	Apply regime flip to raw signal direction	filtered_signal (+1 / -1 / 0)
11	Count consecutive bars of same signal	sig_streak
12	$\text{sig_streak} \geq \text{confirm_bars}$?	confirmed_signal — entry trigger
13	Set entry, $\text{SL} = \text{entry} \pm \text{ATR} \times \text{mult}$, $\text{TP} = \text{SL} \times \text{R}:\text{R}$	Trade opened
14	Monitor price vs TP / SL each bar	[v TP HIT] or [x SL STOP]
15	confirmed_signal reverses or goes flat	[EXIT] — signal close

9. Tuning Tips

- **More signals:** Lower sig_mult (try 1.0) and/or reduce confirm_bars to 1. Expect more whipsaw but faster entries.
- **Fewer, cleaner signals:** Raise sig_mult to 2.0+ and confirm_bars to 3. Fewer trades but higher confidence per trade.
- **Tighter stops:** Lower sl_atr_mult (try 1.0). More frequent stops but smaller loss per stop.
- **Higher timeframe:** The strategy works on any timeframe. On 15-min or 1-hour charts, the signals are less frequent and the ATR-based stops naturally widen.
- **Different R:R:** A 3:1 rr_ratio means fewer TP hits but each win is larger. Balance against win rate shown in the stats table.
- **Regime filter only:** Set sig_mult to 0 to disable the dislocation filter and trade every regime-confirmed bar — useful for studying regime behaviour alone.